

Class Field Theory

1 0908

Goal Describe all abelian extension of number fields.
i.e. inverse Galois problem for abelian groups.

Recall (Kronecker-Weber Theorem) Every abelian extension of the rational numbers is contained in a cyclotomic field.

What happens if we replace $\mathbb{Q}$ by a number field K ?

Denote $\text{Cl}(K)$ as the ideal class group of K . (Fractional ideals modulo principal ideals.)

Theorem 1.1: Let L be the maximal unramified abelian extension of a number field K . Then L/K is finite and $\text{Gal}(L/K) \cong \text{Cl}(K)$.
 L is called the Hilbert class field of K .

Recall Let $N > 1$ be an integer. ζ_N denotes a primitive N -th root of unity. Then

1. $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.
2. Suppose $N \not\equiv 2 \pmod{4}$. Then p is ramified in $\mathbb{Q}(\zeta_N)$ if and only if $p \mid N$.
3. If p splits completely in $\mathbb{Q}(\zeta_N)$, then p is unramified, so the decomposition group is isomorphic to $\text{Gal}(\kappa(P)/\kappa(p))$.

The Frobenius element $\text{Frob}(P/p)$ is the unique element in the decomposition group that acts as $x \mapsto x^p$ on the residue field.

It is too restrictive to ask for unramified extensions.

So we ask for abelian extensions.

1.1 Review $\mathbb{A}_K, \mathbb{I}_K = \mathbb{A}_K^\times, \text{Cl}(K)$

Let I be a fractional ideal in K . Then I can be written as $I = P_1^{a_1} \dots P_r^{a_r}$, where P_i are prime ideals and $a_i \in \mathbb{Z}$.

So the group of fractional ideals is a free abelian group generated by the prime ideals.

$$\bigoplus_P \mathbb{Z}$$

and Cl is the quotient of $\bigoplus_P \mathbb{Z}$ by the image of the map $K^\times \rightarrow \bigoplus_P \mathbb{Z}$ given by $x \mapsto (v_P(x))_P$, where v_P is the valuation at P .

Let $U = \prod_{v:\text{inf}} K_v^\times \times \prod_{v:\text{fin}} \mathcal{O}_v^\times$. It is an open subgroup of $\mathbb{A}_K^\times$.

Then $\mathbb{A}_K^\times/U$ is exactly $\bigoplus_{v:\text{fin}} \mathbb{Z}$. So we have a surjective map $\mathbb{A}_K^\times \rightarrow \text{Cl}(K)$, and the kernel is $UK^\times$.

As the class group corresponds to the maximal unramified abelian extension, we want to shrink U to a smaller open subgroup to detect more abelian extensions.

[Ray class group]@Obsidian

Remark: This isomorphism shows that $\text{Cl}(K, \mathfrak{a})$ is finite:

By the isomorphism, it is discrete.

Also, it is a quotient of $\mathbb{A}_K^1/K^\times$, which is compact, so it is finite.

Example 1.2: $\text{Cl}(\mathbb{Q}, N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.

[計算 $\text{Cl}(\mathbb{Q}, N)$]@Obsidian

By Kronecker-Weber, we know that $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ is the inverse limit of $(\mathbb{Z}/N\mathbb{Z})^\times$ over N .

By above, $(\mathbb{Z}/N\mathbb{Z})^\times$ is a quotient of $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$.

Class field theory is saying that we can construct a map from $C_K = \mathbb{A}_K^\times/K^\times$ to $\text{Gal}(K^{\text{ab}}/K)$ for any number field K , and the image of this map is dense.

Theorem 1.3 (Local class field theory): Let K be a local field. There is a unique continuous homomorphism

$$\rho_K : K^\times \rightarrow \text{Gal}(K^{\text{ab}}/K)$$

satisfying the following properties:

1. For any L/K finite abelian extension, the kernel of ρ_K is exactly $N_{L/K}(L^\times)$, the norm of $L^\times$ in $K^\times$.

$$\begin{array}{ccccc} K^\times & \longrightarrow & \text{Gal}(K^{\text{ab}}/K) & \twoheadrightarrow & \text{Gal}(K^{\text{un}}/K) \\ \downarrow & & \downarrow & \swarrow & \\ \mathbb{Z} & \longrightarrow & \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) & & \end{array}$$

- 2.
3. There is a 1-1 correspondence between U : an open subgroup of $\text{Gal}(K^{\text{ab}}/K)$ and V : open subgroup of finite index of $K^\times$ given by $U \mapsto \rho_K^{-1}(U)$.

2 0910

Definition (CSA): A (finite-dimensional) central simple algebra (CSA) over a field K is a finite-dimensional K -algebra R such that:

- R is a finite-dimensional K -algebra.
- R is central: The center of R is exactly K .
- R is simple: R has no nontrivial two-sided ideals.

Example 2.1: The matrix algebra $M_{n(K)}$ is a CSA over K .

Fact 2.2 (Wedderburn): Let A be a CSA over K . Then there's a central division algebra D s.t.

$$A \cong M_n(D)$$

for some $n \in \mathbb{Z}_{>0}$ and D is unique up to isomorphism.

Definition Let A, B be two CSAs over K . We say A and B are Brauer equivalent if there exist $m, n \in \mathbb{Z}_{>0}$ s.t.

$$A \otimes_K M_m(K) \cong B \otimes_K M_n(K)$$

i.e. their corresponding division algebras are isomorphic. The Brauer equivalence class of A is denoted by $[A]$.

The Brauer group $\text{Br}(K)$ is the set of Brauer equivalence classes of CSAs over K with the group operation given by tensor product over K .

Then $[A]^{-1} = [A^{\text{op}}]$.

The importance of the Brauer group in CFT:

If K_v is non-archimedean, then $\text{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}$ and it is canonical.

We can write $\text{Br}(K_v)$ explicitly as its element is the form (χ, a) where χ is a character of $\text{Gal}(K_v^{\text{ab}}/K_v)$ and $a \in K_v^\times$.

It help us define a map

$$\begin{aligned} \rho : K_v^\times &\rightarrow \text{Hom}(\text{Hom}(\text{Gal}(K_v^{\text{ab}}/K_v), \mathbb{Q}/\mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \\ a &\mapsto (\chi \mapsto \text{inv}(a, \chi)) \end{aligned}$$

and the right hand side is $\text{Gal}(K_v^{\text{ab}}/K_v)$ given by Pontryagin duality.

3 Lecture 1: Group cohomology

Let G be a finite group.

Definition

1. M is a G -module if M is an abelian group with a G -action that is compatible with the group structure.

We also consider M as a $\mathbb{Z}[G]$ -module.

2. $f : M \rightarrow N$ is a G -module homomorphism if f is a map preserving the G -action.

Definition $C^n(\mathbb{Z}[G], A) = \text{Hom}(\mathbb{Z}[G]^{n+1}, A)$.

$\partial^n : C^n(\mathbb{Z}[G], A) \rightarrow C^{n+1}(\mathbb{Z}[G], A)$ is defined by

$$(\partial^n f)(g_0, \dots, g_{n+1}) = \sum_{i=0}^{n+1} (-1)^i f(g_0, \dots, \hat{g}_i, \dots, g_{n+1})$$

It defines a cochain complex $(C^n(\mathbb{Z}[G], A), \partial^n)$, and the cohomology of this complex is denoted by $H^n(G, A)$.

4 0917

Definition A G -module B is called relatively injective if B is a direct summand of an induced module $\text{Ind}_1^G(A)$ for some abelian group A .

Corollary 4.1: A relatively injective G -module is acyclic.

5 Lecture 2: Change of groups

General: $f : G' \rightarrow G$: group homomorphism of finite groups.

Let A be a G -module. Through f , we can make A a G' -module. Denote as f^*A . This is a functor $f^* : G\text{-Mod} \rightarrow G'\text{-Mod}$. We have a natural map $f^* : H^n(G, A) \rightarrow H^n(G', f^*A)$ for all n .

Now consider the case $H \hookrightarrow G$. The pullback map is called the restriction map.

Theorem 5.1 (Shapiro lemma):

$$H^n(H, A) \cong H^n(G, \text{Ind}_H^G(A))$$

Proof: By Frobenius reciprocity, we have

$$\text{Hom}_G(B, \text{Ind}_H^G(A)) \cong \text{Hom}_H(B, A)$$

For $n = 0$: Take $B = \mathbb{Z}$ with trivial action.

Now for $n > 0$, take an injective resolution $A \rightarrow I^\bullet$. By Frobenius reciprocity and the exactness of induction functor, $\text{Ind}_H^G(I^\bullet)$ is an injective resolution of $\text{Ind}_H^G(A)$.

Applying the case $n = 0$ to $I^\bullet$, we get the desired isomorphism since both sides are universal δ -functors.

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Consider another case $H \hookrightarrow G$ and A is a G -module. Consider the following map:

For $n = 0$, we have a map $H^0(H, A|_H) \rightarrow H^0(G, A)$ given by $a \mapsto \sum_{g \in G/H} ga$.

Now, $H^n(H, A|_H)$ is a δ -functor. We want to show that it's the derived functor of $A \mapsto A^H =$

$H^0(H, A|_H)$.

By the dual of Frobenius reciprocity,

$$\mathrm{Hom}_H(- \otimes_H G, I) \cong \mathrm{Hom}_G(-, I)$$

Since $- \otimes_H G$ is exact, $I^\bullet|_H$ is an injective resolution of $A|_H$. (Restriction is exact.)

This proves that $H^n(H, A|_H)$ is the derived functor of $A \mapsto A^H$ hence a universal δ -functor.

As $A \mapsto H^n(G, A)$ is also a δ -functor, we have a natural transformation $H^n(H, A|_H) \rightarrow H^n(G, A)$ for all n .